

Phạm Khánh Minh Mẫn

Backend Engineer · E-commerce API Integration · LLM-Agent Memory Research

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SUMMARY

Backend engineer with 5+ years in e-commerce and multi-marketplace API integration. Focused on NestJS, TypeScript, and event-driven systems on AWS.

Author of several open-source libraries for the Shopee, TikTok Shop and Lazada Open APIs - organised as a monorepo with npm workspaces, Changesets and automated CI/CD.

Hands-on with order processing, inventory sync, recurring billing on Stripe and PayPal, local payment gateways (Fundii, Payoo, ZaloPay), shipping providers (GHN, Ahamove, TikiNOW) and ERP (NaviWorld).

Since 2026, a graduate student in Computer Science (research track) at Danang University of Science and Technology.

Research interest: memory for LLM agents - specifically, separating what graph structure contributes from what the data representation contributes, using budget-matched controlled experiments and statistics at the correct unit of analysis.

SELECTED SYSTEMS

Multi-environment deploy pipeline

Backend infrastructure

Dev / staging / production deploy path for a four-service product, designed and built end to end.

- One workflow per environment; production takes a service argument so you deploy one service, not all four.
- Images built with Buildx layer cache, tagged by commit SHA, pushed to ECR.
- Deploy registers a new ECS task definition, updates the service, and waits for it to stabilise.

GitHub Actions · Docker Buildx · AWS ECR · AWS ECS · ALB · Docker Compose

BattleCatsLab battlecatslab.fun

Personal project · public

Data and simulation site for a mobile game - 9,000 generated pages, a deterministic battle engine, and server-authoritative realtime PvP.

- Server owns the sim and ticks it at 30 Hz; clients send deploy intents and render snapshots, so a client cannot fake a result.
- Ranked ELO gate treats same-IP as telemetry only - shared Wi-Fi must not invalidate a real match; the abuse lockout derives from an append-only log, not a mutable counter.
- A golden-vector test pins fixed team pairs to their exact winner and frame count, so an engine change that would desync stored replays fails CI instead of corrupting them silently.

TypeScript · React · Vite · SQLite · Cloudflare Workers · Cloudflare R2 · SSE

Virtual factory / production scheduling

Manufacturing simulation · client project

Django platform simulating a factory floor - scheduling, facility state, dispatching - bridged to real equipment over OPC-UA.

- Five services share one models package: any schema change ripples through all of them.
- Traced the DB -> data-creator -> environment path and reported where the schema had already drifted apart.

Python · Django · PostgreSQL · Redis · Docker Compose · OPC-UA · Java

Publishing platform (portal + identity)

Content platform · client project

Four applications behind one account, sharing a single auth package.

- Next.js and Nuxt apps hold the same auth contract - keeping that contract stable is the whole constraint.
- Single sign-on on Cognito, wrapped so no app talks to Cognito directly.

Next.js · Nuxt · TypeScript · AWS Cognito · AWS SES · Algolia · Bugsnag

Subscription billing on Stripe and PayPal

Recurring payments · client project

Rails subscription platform for a Japanese publisher: plan upgrades, webhook settlement and tax-compliant receipts, across two payment providers at once.

- Stripe signs the raw body, so the HMAC verifies locally; PayPal needs an OAuth2 token and a call back to verify-webhook-signature - verification itself is a network hop that can fail.
- Both deliver at least once and out of order: a conditional UPDATE decides which delivery settles the payment, and stale events are dropped by timestamp so a late failure notice cannot revive a cancelled subscription.
- The webhook usually beats the browser back, so payment settles there rather than on the success redirect, with the subscription re-read from the gateway API instead of the event payload.

- Dunning differs too: Stripe marks past_due only after its retries are exhausted, while PayPal walks PAYMENT.FAILED -> SUSPENDED -> CANCELLED.

Ruby on Rails 8 · PostgreSQL · Stripe · PayPal Subscriptions · Sidekiq · Redis · RSpec · Heroku

Aptitude-test study app (backend)

EdTech · client project

FastAPI service on Firestore: question bank, topic taxonomy, per-user progress.

- Rapid mark-difficult / mark-unlearned calls overwrote each other's stats - a write race, not a slow query.
- Fixed by updating counters incrementally instead of recomputing the category, which also dropped a full scan per request.

Python · FastAPI · Firebase Firestore · BigQuery · Cloud Build

EXPERIENCE

Backend Developer · DiproTech

2026 - present

Manufacturing simulation, a subscription publishing platform and a study app, for Japanese clients.

- Engineering detail in Systems above.

Python · Django · FastAPI · Ruby on Rails · PostgreSQL · Firestore · Stripe · PayPal · OPC-UA

Backend Developer - Social App · DiproTech

07/2025 - 03/2026

Newsfeed and content distribution, team of 10.

- Owned the newsfeed backend; designed the seed-based ranking it runs on.
- Built the CI/CD and ECS deploy path the team ships through.

Node.js · TypeScript · PostgreSQL · Redis · AWS ECS · GitHub Actions

Backend Developer - PaymentShield · Devtify Technologies

01/2025 - 06/2025

Auto loan service between lenders, dealers and customers, team of 7.

- Started the codebase: NestJS, Docker, PostgreSQL, layered by responsibility.
- Webhooks publish to SQS and Lambda consumes them, so a slow downstream cannot block the callback.
- Failed jobs retry with backoff and land in a DLQ instead of disappearing.

NestJS · TypeScript · PostgreSQL · AWS Lambda · AWS SQS · BullMQ · Redis

Backend Developer - ROUTINE · Devtify Technologies

01/2024 - 03/2025

Orders, inventory and refunds unified across Shopee, Lazada and TikTok Shop, team of 10.

- One order model over three marketplaces, each with its own API shape and failure modes.
- Kept stock and finance in step with the NaviWorld ERP.
- Integrated three payment gateways and three shipping providers.

NestJS · TypeScript · PostgreSQL · Redis · Elasticsearch · BullMQ · Shopee API · TikTok Shop API · Lazada API

Backend Developer - BEAUTYBOX / THEFACESHOP / REEBOK · Devtify Technologies

04/2022 - 11/2023

Retail digital transformation for HSVGroup, team of 15.

- Designed the database and REST API; unified order processing across three marketplaces.

NestJS · TypeScript · PostgreSQL · Redis · AWS EC2 · Docker

Military Service · Vietnam People's Army

01/2020 - 01/2022

Completed compulsory military service.

Backend Developer - SunWorld B2B Ticket · D-SOFT JSC

03/2019 - 12/2019

Ticket selection, online payment and e-ticket storage, team of 7.

- Designed and built the REST API on PHP / Laravel / SQL Server.

PHP · Laravel · SQL Server

OPEN-SOURCE PACKAGES

Shopee API Client npm: shopee-api-client

TypeScript client for Shopee Open API v2. Covers seller authorization, token management, orders, products, logistics, and payment escrow.

TikTok Shop API Client npm: tiktokshops-api-client

TypeScript API client for TikTok Shop Open API. Covers seller authorization, order APIs, product APIs, and fulfillment APIs.

Lazada API Client npm: lazada-api-client

TypeScript API client for Lazada Open API. Covers seller authorization, order APIs, and product APIs.

All-in-One Package npm: shopee-tiktokshops-lazada-api

Bundles the Shopee, TikTok Shop and Lazada SDKs into a single dependency, with versions kept in sync automatically.

SKILLS

Languages	TypeScript, JavaScript, PHP, Python, Ruby
Backend Frameworks	NestJS, Node.js, Express, Laravel, Ruby on Rails, Django, FastAPI
Databases & Search	PostgreSQL, Redis, Elasticsearch, SQL Server
E-commerce Platforms	Shopee Open API v2, TikTok Shop API v2, Lazada Open Platform, GHN / Ahamove / TikiNOW
Cloud & DevOps	AWS EC2 / S3 / ECS, AWS Lambda / SQS, Docker, GitHub Actions, GitLab CI, Linux
Message Queues & Workers	BullMQ, Redis Queue, AWS SQS + DLQ
Payment Integrations	Stripe (Checkout, Subscriptions, Webhooks), PayPal Subscriptions, Fundiin, Payoo, ZaloPay, Recurring billing & proration, Webhook idempotency, Invoicing & consumption tax (適格請求書)

EDUCATION

M.Sc. in Computer Science <i>Danang University of Science and Technology (DUT) - The University of Danang</i> Research track. Coursework in progress; thesis topic not yet decided.	2026 - present
B.Sc. in Information Technology <i>University of Science and Education - The University of Danang</i> Major in Information Technology.	2015 - 2019

RESEARCH

When Does Graph Memory Help Beyond Verbalized Relations? Isolating Representation from Structure in LLM-Agent Memory

2026 - present | Exploratory - no published results | Independent research - not committed as a thesis topic

Question. Graph memory is widely believed to help LLM agents remember better. But when graph memory is compared against flat memory, existing work changes TWO things at once: structure (the ability to traverse relations) and representation (information rewritten as typed relational sentences - unavoidable, because an LLM can only read text). So which of the two does the observed benefit belong to?

Method. Verbalize-control: three token-budget-matched conditions drawing from the same candidate edge pool and differing only in the selector - graph traversal, cosine truncated to the budget, and dump-everything. Plus a degree-preserving shuffled-edge control to separate “these particular edges” from “merely having a graph”. The design is pre-registered, evaluated on an external non-circular benchmark, and specifies the negative branch in advance (TOST equivalence testing).

Status, stated plainly: this is something I work on outside of client work, not a committed thesis topic. A 30-question feasibility probe has been run; the quality gate fired and returned “inconclusive” rather than a result. No publishable numbers.